

Association between use of interferon beta and progression of disability in patients with relapsing-remitting multiple sclerosis.



CONTEXT: Interferon beta is widely prescribed to treat multiple sclerosis (MS); however, its relationship with disability progression has yet to be established. OBJECTIVE: To investigate the association between interferon beta exposure and disability progression in patients with relapsing-remitting MS. DESIGN, SETTING, AND PATIENTS: Retrospective cohort study based on prospectively collected data (1985-2008) from British Columbia, Canada. Patients with relapsing-remitting MS treated with interferon beta (n = 868) were compared with untreated contemporary (n = 829) and historical (n = 959) cohorts. MAIN OUTCOME MEASURES: The main outcome measure was time from interferon beta treatment eligibility (baseline) to a confirmed and sustained score of 6 (requiring a cane to walk 100 m; confirmed at >150 days with no measurable improvement) on the Expanded Disability Status Scale (EDSS) (range, 0-10, with higher scores indicating higher disability). A multivariable Cox regression model with interferon beta treatment included as a time-varying covariate was used to assess the hazard of disease progression associated with interferon beta treatment. Analyses also included propensity score adjustment to address confounding by indication. RESULTS: The median active follow-up times (first to last EDSS measurement) were as follows: for the interferon beta-treated cohort, 5.1 years (interquartile range [IQR], 3.0-7.0 years); for the contemporary control cohort, 4.0 years (IQR, 2.1-6.4 years); and for the historical control cohort, 10.8 years (IQR, 6.3-14.7 years). The observed outcome rates for reaching a sustained EDSS score of 6 were 10.8%, 5.3%, and 23.1% in the 3 cohorts, respectively. After adjustment for potential baseline confounders (sex, age, disease duration, and EDSS score), exposure to interferon beta was not associated with a statistically significant difference in the hazard of reaching an EDSS score of 6 when either the contemporary control cohort (hazard ratio, 1.30; 95% CI, 0.92-1.83; P = .14) or the historical control cohort (hazard ratio, 0.77; 95% CI, 0.58-1.02; P = .07) were considered. Further adjustment for comorbidities and socioeconomic status, where possible, did not change interpretations, and propensity score adjustment did not substantially change the results. CONCLUSION: Among patients with relapsing-remitting MS, administration of interferon beta was not associated with a reduction in progression of disability.